Coupled Two-Level Atom: Time Evolution

Problem

Consider a two-level atom interacting with a light field described by the Hamiltonian:

$$\hat{H} = \begin{pmatrix} 0 & \frac{1}{2}\hbar\Omega \\ \frac{1}{2}\hbar\Omega & -\hbar\Delta \end{pmatrix}$$

where:

- $\Delta = \omega_0 - \omega$ is the detuning
- Ω is the Rabi frequency

We aim to:

1. Find eigenvalues and eigenvectors
2. Construct the time evolution operator
3. Obtain the time evolution of the state

Step 1: Eigenvalues

Solve the characteristic equation:

$$\det\left(\hat{H} - EI\right) = 0$$

$$\begin{vmatrix} -E & \frac{1}{2}\hbar\Omega \\ \frac{1}{2}\hbar\Omega & -\hbar\Delta - E \end{vmatrix} = 0$$

$$E(E + \hbar\Delta) - \frac{\hbar^2\Omega^2}{4} = 0$$

$$E^2 + \hbar\Delta E - \frac{\hbar^2\Omega^2}{4} = 0$$

Solving:

$$E_{\pm} = \frac{-\hbar\Delta \pm \hbar\Omega_R}{2}$$

where the **generalized Rabi frequency** is:

$$\Omega_R = \sqrt{\Omega^2 + \Delta^2}$$

Step 2: Eigenvectors

Solve:

$$\hat{H} |\psi\rangle = E |\psi\rangle$$

Let $|\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$.

From first row:

$$\frac{1}{2}\hbar\Omega c_2 = E c_1$$

$$c_2 = \frac{2E}{\hbar\Omega} c_1$$

Thus eigenvectors:

$$|E_{\pm}\rangle = \begin{pmatrix} 1 \\ \frac{2E_{\pm}}{\hbar\Omega} \end{pmatrix}$$

These can be normalized if needed.

Step 3: Time Evolution Operator

The time evolution operator is:

$$U(t) = e^{-i\hat{H}t/\hbar}$$

Using spectral decomposition:

$$U(t) = e^{-iE_+t/\hbar} |E_+\rangle \langle E_+| + e^{-iE_-t/\hbar} |E_-\rangle \langle E_-|$$

Step 4: Time Evolution of State

Given initial state $|\psi(0)\rangle$, we expand:

$$|\psi(0)\rangle = a_+ |E_+\rangle + a_- |E_-\rangle$$

Then:

$$|\psi(t)\rangle = a_+ e^{-iE_+t/\hbar} |E_+\rangle + a_- e^{-iE_-t/\hbar} |E_-\rangle$$

Step 5: Explicit Solution (Rabi Oscillations)

Assume initial state $|\psi(0)\rangle = |1\rangle$:

The probability of excitation becomes:

$$P_2(t) = \frac{\Omega^2}{\Omega_R^2} \sin^2\left(\frac{\Omega_R t}{2}\right)$$

This describes **Rabi oscillations**.

Final Result

$$\boxed{P_2(t) = \frac{\Omega^2}{\Omega_R^2} \sin^2\left(\frac{\Omega_R t}{2}\right)}$$

Insights

- $\Omega_R = \sqrt{\Omega^2 + \Delta^2}$ governs oscillation frequency
- On resonance ($\Delta = 0$):
$$P_2(t) = \sin^2\left(\frac{\Omega t}{2}\right)$$
- Off-resonance reduces oscillation amplitude
- Maximum transition probability:
$$P_{\max} = \frac{\Omega^2}{\Omega_R^2}$$

Physical Interpretation

- System behaves like a two-level quantum oscillator
- Light field drives coherent transitions
- Detuning introduces energy mismatch
- Rabi frequency determines coupling strength